

ON A QUESTION OF BÄCHLE ON SOLVABLE CUT GROUPS

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ABSTRACT. A finite group is a *cut group* if all central units of its integral group ring are trivial. Bächle proved that solvable cut groups are $\{2, 3, 5, 7\}$ -groups and asked whether $\exp O_p(G)$ divides p for every solvable cut group G and $p \in \{5, 7\}$. We show that the answer is negative for both primes: the normalizer N_p of a Sylow p -subgroup of the symmetric group S_{p^2} is a solvable cut group whose Sylow p -subgroup is normal, isomorphic to $C_p \wr C_p$, and of exponent p^2 . More precisely, the field generated by the character values of N_p is $\mathbb{Q}(\zeta_{p-1})$ for every prime p , so N_p is a cut group exactly when $p \in \{2, 3, 5, 7\}$. The proof is computer-free; it rests on a criterion for split extensions $V \rtimes H$ in which $Z(H)$ induces all scalars on V . We complement this with some restrictions: a counterexample must satisfy $|O_p(G)| \geq p^3$, there is none of order at most 2000, the analogous construction in degree p^3 (which has a normal Sylow p -subgroup of exponent p^3) does not give a cut group, and the answer is positive for groups of odd order.

1. INTRODUCTION

A finite group G is a *cut group* if every central unit of the integral group ring $\mathbb{Z}G$ is trivial, that is, of the form $\pm z$ with $z \in Z(G)$. By a theorem of Ritter and Sehgal [RS90] this property is purely group-theoretic: G is a cut group if and only if, for every $x \in G$, every generator of $\langle x \rangle$ is conjugate in G to x or to x^{-1} . In the group-theoretic literature such groups are called *inverse semi-rational* [CD10]. In terms of characters, G is a cut group if and only if the field of values $\mathbb{Q}(\chi)$ of every $\chi \in \text{Irr}(G)$ is either $\mathbb{Q}$ or an imaginary quadratic field. Rational groups are therefore cut groups, but the class of cut groups is much larger; see [BMP17, BCJM21] and the recent survey [CM25], and [BKMdR23, DGdR26] for recent work on solvable cut groups.

Gow [Gow76] proved that 2, 3 and 5 are the only primes that can divide the order of a solvable rational group, and Hegedűs [Heg05] proved that the Sylow 5-subgroup of a solvable rational group is normal and elementary abelian. For cut groups, Bächle [Bäc18, Theorem 1.2] proved that the order of a solvable cut group is divisible only by primes in $\{2, 3, 5, 7\}$. In [Bäc19] he observed that, for $p \in \{5, 7\}$, iterated wreath products of $\text{AGL}(1, p) = N_{S_p}(C_p)$ are solvable cut groups with arbitrarily large p -length whose Sylow p -subgroups have arbitrarily large exponent, and he asked whether at least the Hall $\{5, 7\}$ -subgroups of the Fitting layers are well behaved:

Question 1.1 (Bächle [Bäc19, Question 3]). Let G be a solvable cut group. Is it true that $\exp O_p(G)$ divides p for $p \in \{5, 7\}$?

This question also appears in the Kourovka Notebook as Problem 20.4(c). The purpose of this note is to show that the answer is negative for both primes. The counterexamples are very natural ones: Sylow normalizers in S_{p^2} .

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Theorem A. *Let p be a prime, let P be a Sylow p -subgroup of the symmetric group S_{p^2} , and let $N_p = N_{S_{p^2}}(P)$. Then:*

- (a) N_p is solvable of order $p^{p+1}(p-1)^2$, and its normal Sylow p -subgroup $P = O_p(N_p)$ is isomorphic to $C_p \wr C_p$ and has exponent p^2 ;
- (b) $\mathbb{Q}(N_p) = \mathbb{Q}(\zeta_{p-1})$, where $\mathbb{Q}(N_p)$ is the field generated by the values of all irreducible characters of N_p ;
- (c) N_p is a cut group if and only if $p \in \{2, 3, 5, 7\}$, and it is a rational group if and only if $p \in \{2, 3\}$.

Corollary 1.2. *For $p \in \{5, 7\}$, the group N_p is a solvable cut group with $\exp O_p(N_p) = p^2$. In particular, [Question 1.1](#) has a negative answer, both for $p = 5$ and for $p = 7$.*

Explicitly, N_5 has order $2^4 \cdot 5^6 = 250\,000$ and $\mathbb{Q}(N_5) = \mathbb{Q}(i)$, while N_7 has order $2^2 \cdot 3^2 \cdot 7^8 = 207\,532\,836$ and $\mathbb{Q}(N_7) = \mathbb{Q}(\sqrt{-3})$.

The proof of [Theorem A](#), given in [Sections 3](#) and [4](#), is short and does not use a computer. The group N_p is the split extension of the permutation module $V = \mathbb{F}_p^{\mathbb{F}_p}$ of $\text{AGL}(1, p)$ by the group $\mathbb{F}_p^\times \times \text{AGL}(1, p)$, where $\mathbb{F}_p^\times$ acts by scalars ([Proposition 4.4](#)). The key tool is [Proposition 3.1](#): if $Z(H)$ induces all scalar transformations on an $\mathbb{F}_p H$ -module V , then the character fields of $V \rtimes H$ are exactly the character fields of the stabilizers in H of the linear characters of V . For N_p these stabilizers are $C_{p-1} \times \text{AGL}(1, p)$, $\text{AGL}(1, p)$, and cyclic groups of order dividing $p-1$ ([Lemma 4.2](#)). We have also confirmed [Theorem A](#) for $p \in \{2, 3, 5, 7\}$ by direct computer calculation ([Section 7](#)). The group N_p is a subgroup of Bächle's wreath product $\text{AGL}(1, p) \wr \text{AGL}(1, p)$ ([Remark 4.5](#)).

In [Sections 5](#) and [6](#) we collect some further information on [Question 1.1](#).

- If $p \geq 5$, a cut group has no normal cyclic subgroup of order p^2 ([Proposition 5.1](#)). Hence a solvable cut group G with $\exp O_p(G) > p$ satisfies $|O_p(G)| \geq p^3$, and a computer search shows that there is no such group of order at most 2000 ([Proposition 5.2](#)).
- The natural next candidate, a subgroup $W_p \leq S_{p^3}$ built in the same way and having a normal Sylow p -subgroup of exponent p^3 , is *not* a cut group for $p \geq 5$ ([Proposition 5.4](#)); for $p = 5$, W_5 is the Sylow normalizer of S_{125} . We do not know whether $\exp O_p(G)$ is bounded for solvable cut groups ([Question 5.5](#)).
- For groups of odd order the answer to [Question 1.1](#) is positive ([Theorem B](#)). This is also a consequence of a theorem of Maheshwary [[Mah18](#), Theorem 1]; we give a short self-contained proof.

Theorem B. *Let G be a cut group of odd order. Then $5 \nmid |G|$, and every element of G of 7-power order has order 1 or 7. In particular $\exp O_5(G) = 1$ and $\exp O_7(G)$ divides 7.*

Notation 1.3. All groups are finite. For $x \in G$ we write $o(x)$ for the order of x and $B_G(x) = N_G(\langle x \rangle)/C_G(x)$. We write ζ_n for a primitive complex n -th root of unity, φ for Euler's function, $\mathbb{Q}(\chi)$ for the field of values of a character χ , and $\mathbb{Q}(G)$ for the field generated by $\mathbb{Q}(\chi)$, $\chi \in \text{Irr}(G)$. Groups act on the left. For a class function θ of a subgroup $I \leq G$ and $g \in N_G(I)$, ${}^g\theta(y) = \theta(g^{-1}yg)$. The Galois group $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ acts on class functions with values in $\mathbb{Q}(\zeta_n)$ by $\theta^\sigma = \sigma \circ \theta$. Commutators are $[x, y] = x^{-1}y^{-1}xy$. Finally, $C_p \wr C_p$ denotes the regular wreath product.

AI disclosure and authorship status. I, the person whose name appears between question marks at the page headers of this document, do not claim to be the author of anything, neither the mathematical ideas nor the redaction, in this paper. It was all generated by Claude Opus 5.5. I merely provided the preprint [[Bäc19](#)] with the question. This said, as the first human

who actually knew this answer to the question and understood it, I may claim some credit for it: I checked with my own GAP code that both N_5 and N_7 are actually cut groups, being the remaining properties for them to be counterexamples to Bächle's question obvious, given the fact that N_p is the normalizer of a Sylow p -subgroup of S_{p^2} . Summarizing, the answer to Bächle's question could actually be two lines of text plus a few more lines of GAP code; thus, the rest of the paper, I didn't read.

2. PRELIMINARIES

2.1. Cut groups. Let $x \in G$ with $o(x) = n$. Conjugation induces an injective homomorphism $B_G(x) \rightarrow \text{Aut}(\langle x \rangle) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, and we identify $B_G(x)$ with its image, which is $\{r \in (\mathbb{Z}/n\mathbb{Z})^\times : x^r \text{ is conjugate to } x\}$. The element x is *inverse semi-rational* if every generator of $\langle x \rangle$ is conjugate to x or to x^{-1} , i.e. if $\langle B_G(x), -1 \rangle = (\mathbb{Z}/n\mathbb{Z})^\times$. Equivalently, either $B_G(x) = (\mathbb{Z}/n\mathbb{Z})^\times$, or $B_G(x)$ has index 2 in $(\mathbb{Z}/n\mathbb{Z})^\times$ and $-1 \notin B_G(x)$ (that is, x is not real). In particular

$$(2.1) \quad x \text{ inverse semi-rational} \implies |B_G(x)| \geq \varphi(o(x))/2.$$

Proposition 2.1 ([RS90]; see also [BMP17, CD10, BCJM21]). *For a finite group G the following are equivalent:*

- (i) G is a cut group;
- (ii) every element of G is inverse semi-rational;
- (iii) for every $\chi \in \text{Irr}(G)$, the field $\mathbb{Q}(\chi)$ is $\mathbb{Q}$ or an imaginary quadratic field.

Lemma 2.2.

- (a) Quotients of cut groups are cut groups.
- (b) An abelian group is a cut group if and only if its exponent divides 4 or 6. In particular a cyclic group of order n is a cut group if and only if $n \in \{1, 2, 3, 4, 6\}$.
- (c) If $\mathbb{Q}(G) \subseteq K$, where K is $\mathbb{Q}$ or an imaginary quadratic field, then G is a cut group.

Proof. (a) $\text{Irr}(G/N) \subseteq \text{Irr}(G)$ by inflation; apply Proposition 2.1(iii). (b) If G is abelian then $B_G(x) = 1$ for all x , so by Proposition 2.1(ii) G is a cut group if and only if $\varphi(o(x)) \leq 2$, i.e. $o(x) \in \{1, 2, 3, 4, 6\}$, for all $x \in G$. As G is abelian, this holds if and only if $\exp G$ divides 4 or 6. (c) Every subfield of K is $\mathbb{Q}$ or K ; apply Proposition 2.1(iii). $\square$

2.2. Characters of split extensions with abelian kernel. The following is the classical description of the irreducible characters of $V \rtimes H$ with V abelian; see [Ser77, §8.2, Proposition 25] or [Isa76, (6.11), (6.17)].

Lemma 2.3. *Let $G = V \rtimes H$ with V abelian. For $\lambda \in \text{Irr}(V)$ let $H_\lambda = \{h \in H : {}^h\lambda = \lambda\}$ and $I_\lambda = VH_\lambda$, the inertia group of λ in G .*

- (a) *The map $\tilde{\lambda}: I_\lambda \rightarrow \mathbb{C}^\times$, $\tilde{\lambda}(vh) = \lambda(v)$ ($v \in V$, $h \in H_\lambda$), is a linear character extending λ .*
- (b) *For $\psi \in \text{Irr}(H_\lambda)$, regarded as a character of I_λ via $I_\lambda/V \cong H_\lambda$, the character $\chi_{\lambda, \psi} = (\tilde{\lambda}\psi)^G$ is irreducible.*
- (c) *Every irreducible character of G is of the form $\chi_{\lambda, \psi}$, and for fixed λ the map $\psi \mapsto \chi_{\lambda, \psi}$ is injective on $\text{Irr}(H_\lambda)$.*

3. A CRITERION FOR SPLIT EXTENSIONS

Proposition 3.1. *Let p be a prime, H a finite group, V a finite $\mathbb{F}_p H$ -module and $G = V \rtimes H$. Suppose that there is a subgroup $Z \leq Z(H)$ whose image in $\text{GL}(V)$ is the group $\mathbb{F}_p^\times \cdot \text{id}_V$ of all*

nonzero scalars. Then, in the notation of [Lemma 2.3](#),

$$\mathbb{Q}(\chi_{\lambda,\psi}) = \mathbb{Q}(\psi) \quad \text{for all } \lambda \in \text{Irr}(V) \text{ and } \psi \in \text{Irr}(H_\lambda).$$

Consequently $\{\mathbb{Q}(\chi) : \chi \in \text{Irr}(G)\} = \bigcup_{\lambda \in \text{Irr}(V)} \{\mathbb{Q}(\psi) : \psi \in \text{Irr}(H_\lambda)\}$, the field $\mathbb{Q}(G)$ is the compositum of the fields $\mathbb{Q}(H_\lambda)$, and G is a cut group if and only if H_λ is a cut group for every $\lambda \in \text{Irr}(V)$.

Proof. We write V additively when convenient; $h \in H$ acts on V by $h \cdot v = hvh^{-1}$. Let $n = |G|$ and $\sigma \in \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$, and let $r \in \mathbb{Z}$ with $\sigma(\zeta_p) = \zeta_p^r$, so $p \nmid r$. By hypothesis there is $z \in Z$ with $z \cdot v = r'v$ for all $v \in V$, where $rr' \equiv 1 \pmod{p}$. Then $z^{-1}vz = rv$, so for $\lambda \in \text{Irr}(V)$, whose values are p -th roots of unity,

$$z\lambda(v) = \lambda(z^{-1}vz) = \lambda(v)^r = \sigma(\lambda(v)), \quad \text{i.e.} \quad z\lambda = \lambda^\sigma.$$

Fix $\lambda \in \text{Irr}(V)$ and $\psi \in \text{Irr}(H_\lambda)$ and put $\theta = \tilde{\lambda}\psi \in \text{Irr}(I_\lambda)$. Since $z \in Z(H)$ centralizes H_λ and normalizes V , it normalizes I_λ , and for $v \in V$, $h \in H_\lambda$ we have $z^{-1}(vh)z = (z^{-1}vz)h$. Hence

$$z(\tilde{\lambda}\psi^\sigma)(vh) = \lambda(z^{-1}vz)\psi^\sigma(h) = \lambda^\sigma(v)\psi^\sigma(h) = \theta^\sigma(vh),$$

so $\theta^\sigma = z(\tilde{\lambda}\psi^\sigma)$. Since induction commutes with σ and G -conjugate characters of I_λ induce to the same character of G ,

$$\chi_{\lambda,\psi}^\sigma = (\theta^\sigma)^G = (z(\tilde{\lambda}\psi^\sigma))^G = (\tilde{\lambda}\psi^\sigma)^G = \chi_{\lambda,\psi^\sigma}.$$

By [Lemma 2.3\(c\)](#), $\chi_{\lambda,\psi}^\sigma = \chi_{\lambda,\psi}$ if and only if $\psi^\sigma = \psi$. As $\mathbb{Q}(\chi_{\lambda,\psi})$ and $\mathbb{Q}(\psi)$ are the fixed fields in $\mathbb{Q}(\zeta_n)$ of $\{\sigma : \chi_{\lambda,\psi}^\sigma = \chi_{\lambda,\psi}\}$ and $\{\sigma : \psi^\sigma = \psi\}$ respectively, they coincide. The remaining statements follow from [Lemma 2.3\(c\)](#) and [Proposition 2.1](#). $\square$

Remark 3.2. (1) For $p = 2$ the hypothesis on Z is empty, and for $p = 3$ it says that some central element of H acts as -1 on V . (2) Some hypothesis of this kind is necessary: for $V = \mathbb{F}_5$ and $H = 1$ the only stabilizer $H_\lambda = 1$ is a cut group, but $G = C_5$ is not. (3) [Proposition 3.1](#) reduces the cut property of $V \rtimes H$ to that of the stabilizers H_λ , which are typically much smaller than G . This makes it a convenient tool for constructing cut groups with a prescribed p -structure.

4. THE CONSTRUCTION AND THE PROOF OF THEOREM A

4.1. Set-up. Fix a prime p and write $\mathbb{F} = \mathbb{F}_p$. Let

$$A = \text{AGL}(1, p) = \{a_{m,b} : m \in \mathbb{F}^\times, b \in \mathbb{F}\}, \quad a_{m,b}(x) = mx + b,$$

acting on $\mathbb{F}$, and let $T = \{\tau_b = a_{1,b} : b \in \mathbb{F}\} \cong C_p$ be its normal subgroup of translations, so $A/T \cong \mathbb{F}^\times \cong C_{p-1}$. Let $V = \mathbb{F}^\mathbb{F}$ be the $\mathbb{F}$ -vector space of all functions $f : \mathbb{F} \rightarrow \mathbb{F}$, on which A acts by $(a \cdot f)(x) = f(a^{-1}(x))$, and let $D = \{d_s : s \in \mathbb{F}^\times\} \cong C_{p-1}$ act on V by scalars, $d_s \cdot f = sf$. The two actions commute, so V is an $\mathbb{F}H$ -module for $H = D \times A$, and we define

$$G_p = V \rtimes H = V \rtimes (D \times A).$$

For $x \in \mathbb{F}$ let $\delta_x \in V$ be the indicator function of $\{x\}$, and let $\mathbf{1} \in V$ be the constant function 1. Note that $a \cdot \delta_x = \delta_{a(x)}$ for $a \in A$.

Lemma 4.1.

- (a) G_p is solvable of order $p^{p+1}(p-1)^2$.
- (b) $P := V \rtimes T$ is a normal Sylow p -subgroup of G_p , so $P = O_p(G_p)$, and $P \cong C_p \wr C_p$.
- (c) $\exp P = p^2$; explicitly, $\delta_0\tau_1$ has order p^2 .

Proof. (a) $|V| = p^p$ and $|H| = (p-1) \cdot p(p-1)$; V is abelian and $H = D \times A$ is metabelian. (b) $T \trianglelefteq A$ and D centralizes A , so $T \trianglelefteq H$ and $VT \trianglelefteq G_p$. Moreover $|P| = p^{p+1} = |G_p|_p$. Since T acts regularly on $\mathbb{F}$, V is the regular permutation module of T , and $P \cong C_p \wr C_p$. (c) Since $P/V \cong C_p$ and $\exp V = p$, we have $g^{p^2} = 1$ for all $g \in P$. For $f \in V$ and $\tau \in T$ we have $\tau f \tau^{-1} = \tau \cdot f$, and induction on k gives $(f\tau)^k = (f + \tau \cdot f + \cdots + \tau^{k-1} \cdot f) \tau^k$, where the sum is computed in V . For $f = \delta_0$ and $\tau = \tau_1$ we get, as $\tau_1^p = 1$ and $\tau_1^k \cdot \delta_0 = \delta_k$,

$$(\delta_0 \tau_1)^p = \sum_{k \in \mathbb{F}} \delta_k = \mathbf{1} \neq 0.$$

Hence $\delta_0 \tau_1$ has order p^2 . □

4.2. Stabilizers. The symmetric bilinear form $\langle \phi, f \rangle = \sum_{x \in \mathbb{F}} \phi(x) f(x)$ on V is non-degenerate, so $\phi \mapsto \lambda_\phi$, where $\lambda_\phi(f) = \zeta_p^{\langle \phi, f \rangle}$, is an isomorphism $V \rightarrow \text{Irr}(V)$. The form satisfies $\langle a \cdot \phi, a \cdot f \rangle = \langle \phi, f \rangle$ for $a \in A$, because A permutes $\mathbb{F}$. Hence for $h = (d_s, a) \in H$,

$${}^h \lambda_\phi(f) = \lambda_\phi(h^{-1} \cdot f) = \zeta_p^{\langle \phi, s^{-1} a^{-1} \cdot f \rangle} = \zeta_p^{\langle s^{-1} a \cdot \phi, f \rangle}, \quad \text{i.e.} \quad {}^h \lambda_\phi = \lambda_{s^{-1} a \cdot \phi}.$$

Therefore the stabilizer of λ_ϕ in H is

$$(4.1) \quad H_\phi := H_{\lambda_\phi} = \{(d_s, a) \in D \times A : a \cdot \phi = s\phi\}.$$

Lemma 4.2. *Let $\phi \in V$.*

- (a) *If $\phi = 0$, then $H_\phi = H \cong C_{p-1} \times \text{AGL}(1, p)$.*
- (b) *If ϕ is a nonzero constant function, then $H_\phi = \{1\} \times A \cong \text{AGL}(1, p)$.*
- (c) *If ϕ is not constant, then H_ϕ is cyclic of order dividing $p-1$.*

Proof. (a) is clear. (b) Here $a \cdot \phi = \phi$ for all $a \in A$, so $(d_s, a) \in H_\phi$ if and only if $s\phi = \phi$, i.e. $s = 1$.

(c) The subgroup $K = D \times T$ is normal in H with $H/K \cong A/T \cong C_{p-1}$, so it suffices to show $H_\phi \cap K = 1$. Let $(d_s, \tau_b) \in H_\phi$, so $\tau_b \cdot \phi = s\phi$. Applying this p times and using $\tau_b^p = 1$ and $s^p = s$, we get $\phi = s^p \phi = s\phi$; as $\phi \neq 0$, $s = 1$. Thus $\tau_b \cdot \phi = \phi$, i.e. $\phi(x-b) = \phi(x)$ for all $x \in \mathbb{F}$. If $b \neq 0$, then $x \mapsto x-b$ is a p -cycle on $\mathbb{F}$, and ϕ would be constant. Hence $b = 0$, and $(d_s, \tau_b) = 1$. □

Proposition 4.3. $\mathbb{Q}(G_p) = \mathbb{Q}(\zeta_{p-1})$. *Consequently G_p is a cut group if and only if $p \in \{2, 3, 5, 7\}$, and it is rational if and only if $p \in \{2, 3\}$.*

Proof. Proposition 3.1 applies to $G_p = V \rtimes H$ with $Z = D \leq Z(H)$, so $\mathbb{Q}(G_p)$ is the compositum of the fields $\mathbb{Q}(H_\phi)$, $\phi \in V$. By Lemma 4.2, each H_ϕ is isomorphic to $C_{p-1} \times \text{AGL}(1, p)$, to $\text{AGL}(1, p)$, or to a cyclic group of order dividing $p-1$. The irreducible characters of $\text{AGL}(1, p)$ are the $p-1$ linear characters inflated from $A/T \cong C_{p-1}$, with values in $\mathbb{Q}(\zeta_{p-1})$, and the character μ^A induced from any $1 \neq \mu \in \text{Irr}(T)$, which vanishes off T and takes the value -1 on $T \setminus \{1\}$; in particular $\mathbb{Q}(\text{AGL}(1, p)) \subseteq \mathbb{Q}(\zeta_{p-1})$. The same holds for $C_{p-1} \times \text{AGL}(1, p)$, whose irreducible characters are products, and for cyclic groups of order dividing $p-1$. Hence $\mathbb{Q}(G_p) \subseteq \mathbb{Q}(\zeta_{p-1})$. Conversely, G_p maps onto $G_p/(V \rtimes A) \cong D \cong C_{p-1}$, whose faithful linear characters have field of values $\mathbb{Q}(\zeta_{p-1})$. Thus $\mathbb{Q}(G_p) = \mathbb{Q}(\zeta_{p-1})$.

If $p \in \{2, 3, 5, 7\}$, then $p-1 \in \{1, 2, 4, 6\}$ and $\mathbb{Q}(\zeta_{p-1})$ is $\mathbb{Q}$, $\mathbb{Q}$, $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$, so G_p is a cut group by Lemma 2.2(c), and it is rational for $p \in \{2, 3\}$. If $p \geq 11$, then $p-1 \notin \{1, 2, 3, 4, 6\}$, so $[\mathbb{Q}(\zeta_{p-1}) : \mathbb{Q}] = \varphi(p-1) > 2$, and the faithful linear characters of the quotient C_{p-1} show that G_p is not a cut group. Finally, G_p is rational if and only if $\mathbb{Q}(\zeta_{p-1}) = \mathbb{Q}$, i.e. $p \in \{2, 3\}$. □

4.3. Identification with the Sylow normalizer. Let $\Omega = \mathbb{F} \times \mathbb{F}$, with blocks $B_i = \{i\} \times \mathbb{F}$ for $i \in \mathbb{F}$. For $f \in V$, $s \in \mathbb{F}^\times$ and $a \in A$ define permutations of Ω by

$$\pi(f)(i, j) = (i, j + f(i)), \quad \pi(d_s)(i, j) = (i, sj), \quad \pi(a)(i, j) = (a(i), j).$$

These define homomorphisms on V , D and A ; $\pi(D)$ and $\pi(A)$ commute; and a direct check gives $\pi(a)\pi(f)\pi(a)^{-1} = \pi(a \cdot f)$ and $\pi(d_s)\pi(f)\pi(d_s)^{-1} = \pi(sf)$. Hence π extends to a homomorphism $\pi: G_p \rightarrow \text{Sym}(\Omega)$, and

$$(4.2) \quad \pi(f d_s a)(i, j) = (a(i), sj + f(a(i))) \quad (f \in V, s \in \mathbb{F}^\times, a \in A).$$

If (4.2) is the identity, then $a = 1$ and, varying j , $s = 1$ and $f = 0$; so π is faithful. We identify G_p with $\pi(G_p) \leq \text{Sym}(\Omega) \cong S_{p^2}$.

Proposition 4.4. *$P = V \rtimes T$ is a Sylow p -subgroup of $\text{Sym}(\Omega)$, and $G_p = N_{\text{Sym}(\Omega)}(P)$. In particular $N_p \cong G_p$.*

Proof. By Legendre's formula, $|S_{p^2}|_p = p^{\lfloor p^2/p \rfloor + \lfloor p^2/p^2 \rfloor} = p^{p+1} = |P|$, so P is a Sylow p -subgroup, and $G_p \leq N := N_{\text{Sym}(\Omega)}(P)$ by Lemma 4.1(b). It remains to show $|N| \leq |G_p|$. For $p = 2$, $|S_4 : P| = 3$ and P is not normal in S_4 , so $N = P = G_2$. Let p be odd.

Step 1: V is the unique abelian subgroup of index p in P . Let $g = f\tau_b \in Z(P)$ with $f \in V$. Then $\delta_0 = g\delta_0g^{-1} = \tau_b \cdot \delta_0 = \delta_b$, so $b = 0$, and $g = f$ commutes with τ_1 , so f is constant. Thus $|Z(P)| = p$. If $A_1 \neq V$ were another abelian subgroup of index p , then $P = A_1V$ and $A_1 \cap V \leq Z(P)$, whereas $|A_1 \cap V| = p^{p-1} > p$. Hence V is characteristic in P , and N normalizes V .

Step 2: N acts on the blocks through A . By (4.2), the V -orbits on Ω are the blocks B_i . Since N normalizes V , it permutes the V -orbits; let $\nu: N \rightarrow \text{Sym}(\mathbb{F})$ be the induced action on block indices. By (4.2), $\nu(P) = T$, a regular cyclic subgroup of $\text{Sym}(\mathbb{F})$, and $\nu(N)$ normalizes it. Hence $\nu(N) \leq N_{\text{Sym}(\mathbb{F})}(T) = A$, and $|\nu(N)| \leq p(p-1)$.

Step 3: the kernel. Let $k \in \ker \nu$, say $k(i, j) = (i, \beta_i(j))$ with $\beta_i \in \text{Sym}(\mathbb{F})$. Note that $P \cap \ker \nu = V$. For $f \in V$, the element $k\pi(f)k^{-1}: (i, j) \mapsto (i, \beta_i(\beta_i^{-1}(j) + f(i)))$ lies in $P \cap \ker \nu = V$, so each β_i normalizes the group of translations of $\mathbb{F}$, i.e. $\beta_i(j) = s_i j + c_i$ with $s_i \in \mathbb{F}^\times$, $c_i \in \mathbb{F}$. Moreover $k\pi(\tau_1)k^{-1}\pi(\tau_1)^{-1}: (i, j) \mapsto (i, \beta_i\beta_{i-1}^{-1}(j))$ also lies in $P \cap \ker \nu = V$, so each $\beta_i\beta_{i-1}^{-1}$ is a translation, i.e. $s_i = s_{i-1}$. Thus all s_i equal some s , and $k = \pi(cd_s)$ with $c(i) = c_i$. Hence $|\ker \nu| \leq p^p(p-1)$.

Therefore $|N| \leq p^p(p-1) \cdot p(p-1) = |G_p|$, and $N = G_p$. Since all Sylow p -subgroups of S_{p^2} are conjugate, $N_p \cong G_p$. $\square$

Proof of Theorem A. By Proposition 4.4, $N_p \cong G_p$, with P corresponding to $V \rtimes T$. Now (a) is Lemma 4.1, and (b) and (c) are Proposition 4.3. $\square$

Remark 4.5. By (4.2), $G_p = \{(i, j) \mapsto (a(i), sj + c(i))\}$ is the subgroup of Bächle's wreath product $\text{AGL}(1, p) \wr \text{AGL}(1, p) = \{(i, j) \mapsto (a(i), s_i j + c_i)\}$ consisting of the elements whose base coordinates have a common linear part $s_i = s$. For p odd, the largest normal p -subgroup of the wreath product is the elementary abelian group $\{(i, j) \mapsto (i, j + c_i)\} \cong C_p^p$. Indeed, if it contained an element u acting nontrivially on the blocks, then commutators of u with base elements whose linear part is nontrivial in a single coordinate would be p -elements of the base group with nontrivial linear parts, which is impossible. Passing to the subgroup with a common linear part makes the Sylow p -subgroup $V \rtimes T$ normal, and Lemma 4.2 shows that the cut property survives.

Remark 4.6. For $p = 3$, Theorem A shows that $N_{S_9}(P)$ is a rational group of order 324 whose normal Sylow 3-subgroup has exponent 9. This is consistent with the literature: the restrictive results quoted above concern the primes 5 and 7.

5. RESTRICTIONS ON COUNTEREXAMPLES, AND THE DEGREE p^3 ANALOGUE

Proposition 5.1. *Let $p \geq 5$ be a prime and G a cut group. Then G has no normal cyclic subgroup of order p^2 . Consequently, if $\exp O_p(G) > p$, then $|O_p(G)| \geq p^3$.*

Proof. Suppose $Y = \langle y \rangle \trianglelefteq G$ with $|Y| = p^2$. Then $N_G(Y) = G$ and $B_G(y) = G/C_G(Y)$. On the one hand, $B_G(y)$ is a quotient of G , hence a cut group by Lemma 2.2(a). It is also isomorphic to a subgroup of the cyclic group $(\mathbb{Z}/p^2\mathbb{Z})^\times$ of order $p(p-1)$, so $|B_G(y)| \in \{1, 2, 3, 4, 6\}$ by Lemma 2.2(b), and $p \nmid |B_G(y)|$. On the other hand, $|B_G(y)|$ divides $p(p-1)$ and is at least $p(p-1)/2$ by (2.1), so $|B_G(y)| \in \{p(p-1), p(p-1)/2\}$ is divisible by p . This is a contradiction. For the last statement, note that a p -group of order at most p^2 and exponent greater than p is cyclic of order p^2 , and $O_p(G)$ is normal. $\square$

Proposition 5.2 (computer calculation). *For $p \in \{5, 7\}$ there is no solvable cut group G with $|G| \leq 2000$ and $\exp O_p(G) > p$.*

Proof. By Proposition 5.1, p^3 divides $|G|$. The SmallGroups library contains all groups of order at most 2000 except those of order 1024. We tested all 1700 groups of order $n \leq 2000$ with $125 \mid n$ or $343 \mid n$ (see Section 7). None of them is a solvable cut group with $\exp O_p(G) > p$. $\square$

Remark 5.3. Thus the smallest order of a solvable cut group G with $\exp O_5(G) > 5$ lies between 2001 and $|N_5| = 250\,000$. Moreover, a computer calculation shows that N_5 is minimal among its subgroups containing V : if K runs over representatives of the conjugacy classes of subgroups of $H = D \times A$, then $V \rtimes K$ is a cut group with $\exp O_5(V \rtimes K) > 5$ only for $K = H$.

It is natural to ask whether the construction of Section 4 can be iterated to produce exponent p^3 . The next result shows that the obvious candidate fails.

Let $p \geq 5$, let $\Omega = \mathbb{F} \times \mathbb{F}$ with the action of G_p given by (4.2), and let $V_3 = \mathbb{F}^\Omega$ be the permutation module of G_p on Ω . Let $D_3 = \{e_s : s \in \mathbb{F}^\times\} \cong C_{p-1}$ act on V_3 by scalars, $e_s \cdot f = sf$, and put

$$W_p = V_3 \rtimes (D_3 \times G_p).$$

Exactly as in Section 4.1, W_p acts faithfully on $\Omega \times \mathbb{F}$ by $f(\omega, j) = (\omega, j + f(\omega))$, $e_s(\omega, j) = (\omega, sj)$ and $g(\omega, j) = (g\omega, j)$ for $g \in G_p$.

Proposition 5.4. *Let $p \geq 5$.*

- (a) $P_3 := V_3 \rtimes P$ is a normal Sylow p -subgroup of W_p and a Sylow p -subgroup of $\text{Sym}(\Omega \times \mathbb{F}) \cong S_{p^3}$. In particular P_3 is the iterated wreath product $C_p \wr C_p \wr C_p$, and $\exp P_3 = p^3$.
- (b) W_p is not a cut group.

Proof. (a) P is normal in $D_3 \times G_p$, so $P_3 \trianglelefteq W_p$, and $|P_3| = p^{p^2} \cdot p^{p+1} = |S_{p^3}|_p$ by Legendre's formula. Every Sylow p -subgroup of S_{p^3} contains a p^3 -cycle, so $\exp P_3 = p^3$.

(b) Proposition 3.1 applies to $W_p = V_3 \rtimes (D_3 \times G_p)$ with $Z = D_3$. Identifying $\text{Irr}(V_3)$ with V_3 through the form $\langle \phi, f \rangle = \sum_{\omega \in \Omega} \phi(\omega)f(\omega)$, the computation leading to (4.1) shows that the stabilizer of λ_ϕ in $D_3 \times G_p$ is $\{(e_s, g) : g \cdot \phi = s\phi\}$, where $(g \cdot \phi)(\omega) = \phi(g^{-1}\omega)$. Take $\phi = \delta_{(0,0)} + \delta_{(0,1)}$. Then $g \cdot \phi = \delta_{g(0,0)} + \delta_{g(0,1)}$ takes the values 0 and 1, so $g \cdot \phi = s\phi$ forces $s = 1$ and $g\{(0,0), (0,1)\} = \{(0,0), (0,1)\}$. Hence the stabilizer of λ_ϕ is isomorphic to

$$S = \text{Stab}_{G_p}(\{(0,0), (0,1)\}),$$

and by Proposition 3.1 it suffices to show that S is not a cut group.

By (4.2), $g = f'd_s a$ lies in S if and only if $a(0) = 0$ and $\{f'(0), s + f'(0)\} = \{0, 1\}$, i.e. $(s, f'(0)) \in \{(1,0), (-1,1)\}$. In particular $|S| = 2(p-1)p^{p-1}$. Let $x = \delta_1 \in V$; then $x \in S$. For $g = f'd_s a \in G_p$ we have $g x g^{-1} = s(a \cdot \delta_1) = s\delta_{a(1)}$, as V is abelian. If $g \in S$

normalizes $\langle x \rangle = \mathbb{F}\delta_1$, then $a(1) = 1$ and $a(0) = 0$, so $a = 1$ and $gxg^{-1} = x^s$ with $s = \pm 1$. Conversely, $g = \delta_0 d_{-1} \in S$ inverts x . Hence $B_S(x) = \{\pm 1\} \leq (\mathbb{Z}/p\mathbb{Z})^\times$, and since $p \geq 5$, $\langle B_S(x), -1 \rangle = \{\pm 1\} \neq (\mathbb{Z}/p\mathbb{Z})^\times$. So x is not inverse semi-rational in S , S is not a cut group, and neither is W_p . $\square$

Since $P_3 \leq W_p$, the group W_p is contained in the Sylow normalizer $N_{S_{p^3}}(P_3)$. For $p = 5$ we checked with GAP that $W_5 = N_{S_{125}}(P_3)$, and that $|S| = 5000$ and S is not a cut group. We expect that W_p is the Sylow normalizer of S_{p^3} for every p .

Question 5.5. Let $p \in \{5, 7\}$. Is there a constant c such that $\exp O_p(G) \leq p^c$ for every solvable cut group G ? In particular, is $\exp O_p(G) \leq p^2$ for every solvable cut group G ?

6. GROUPS OF ODD ORDER: PROOF OF THEOREM B

Theorem B is also a consequence of [Mah18, Theorem 1], which states that every element of a cut group of odd order has order 7 or a power of 3 (see also [CM25]). We include a short self-contained proof of the part relevant to [Question 1.1](#). It uses the Gruenberg–Kegel theorem: a solvable group whose prime graph is disconnected is a Frobenius group or a 2-Frobenius group (see [Wil81]). We also use standard facts on Frobenius groups [Hup67, Kap. V, §8]: the kernel is nilpotent [Tho59], the complement acts fixed-point-freely on the kernel, and Sylow subgroups of complements of odd order are cyclic. In addition, if H is a Frobenius complement in G and $1 \neq X \leq H$, then $N_G(X) \leq H$, because $H \cap H^g = 1$ for $g \notin H$.

Throughout, G is a cut group of odd order. If $x \in G$ has order $n \geq 3$, then $B_G(x)$ has odd order, so $-1 \notin B_G(x)$, and [Proposition 2.1\(ii\)](#) forces

$$(6.1) \quad |B_G(x)| = \varphi(n)/2.$$

Proof of Theorem B. Step 0: $\pi(G) \subseteq \{3, 7\}$. G is solvable by the Feit–Thompson theorem [FT63], so $\pi(G) \subseteq \{2, 3, 5, 7\}$ by [Bäc18, Theorem 1.2]. Clearly $2 \notin \pi(G)$. If $x \in G$ had order 5, then by (6.1) $|B_G(x)| = 2$, which is impossible in a group of odd order. So $\pi(G) \subseteq \{3, 7\}$, and we may assume $7 \mid |G|$.

Step 1: G is a Frobenius or 2-Frobenius group. The group $(\mathbb{Z}/21\mathbb{Z})^\times \cong C_2 \times C_6$ has no subgroup of odd order greater than 3, so by (6.1) G has no element of order 21. If G were a 7-group, then for $x \in G$ of order 7 the group $B_G(x)$ would be a 7-subgroup of $(\mathbb{Z}/7\mathbb{Z})^\times \cong C_6$, hence trivial, contradicting (6.1). Thus $\pi(G) = \{3, 7\}$ and the prime graph of G has no edges. By the Gruenberg–Kegel theorem, G is a Frobenius group or a 2-Frobenius group.

Step 2: the 2-Frobenius case. Let $1 < K < L < G$ be normal subgroups such that L is a Frobenius group with kernel K and G/K is a Frobenius group with kernel L/K . Then L/K is isomorphic to a Frobenius complement of odd order, so its Sylow subgroups are cyclic, and it is nilpotent as a Frobenius kernel. Hence L/K is cyclic. Since $\pi(K) \cap \pi(L/K) = \emptyset = \pi(L/K) \cap \pi(G/L)$ and $\pi(G) = \{3, 7\}$, the group L/K is a q -group and $K, G/L$ are r -groups, where $\{q, r\} = \{3, 7\}$. The group G/L acts faithfully on L/K . If $q = 3$, then the 7-group G/L would embed into $\text{Aut}(C_{3^m})$, which has order $2 \cdot 3^{m-1}$; this is impossible. So $L/K \cong C_{7^m}$ is isomorphic to a Sylow 7-subgroup of G , and G/L is a 3-group embedding into the cyclic group $\text{Aut}(C_{7^m})$ of order $6 \cdot 7^{m-1}$, so $G/L \cong C_3$. Let $\bar{y}$ be a generator of L/K in $\bar{G} = G/K$, which is a cut group by [Lemma 2.2\(a\)](#). As L/K is abelian and normal in $\bar{G}$, the group $B_{\bar{G}}(\bar{y})$ is a quotient of $\bar{G}/(L/K) \cong C_3$. By (6.1), $3 \geq \varphi(7^m)/2 = 3 \cdot 7^{m-1}$, so $m = 1$. Thus the Sylow 7-subgroups of G have order 7.

Step 3: the Frobenius case. Let $G = K \rtimes H$ with Frobenius kernel K and complement H . Since $|K|$ and $|H|$ are coprime, one of them is a 7-group and the other a 3-group. If H were a 7-group, then for $x \in H$ of order 7 the group $B_G(x)$ would be a section of H , because

$N_G(\langle x \rangle) \leq H$. It would therefore be a 7-subgroup of $(\mathbb{Z}/7\mathbb{Z})^\times$, hence trivial, contradicting (6.1). So H is a 3-group, cyclic as an odd-order Frobenius complement, and K is the normal Sylow 7-subgroup of G . If h generates H , then $N_G(\langle h \rangle) \leq H \leq C_G(h)$, so $B_G(h) = 1$ and (6.1) gives $\varphi(|H|) = 2$, i.e. $|H| = 3$. Thus $H = \langle h \rangle \cong C_3$ acts fixed-point-freely on K .

By a theorem of Burnside and B.H. Neumann [Neu56], a group with a fixed-point-free automorphism of order 3 is nilpotent of class at most 2. As 7 is odd, the Baer correspondence [Bae38] provides a Lie ring structure $(K, +, [\cdot, \cdot])$ on the set K such that:

- (B1) the Lie bracket coincides with the group commutator;
- (B2) $xy = x + y + \frac{1}{2}[x, y]$ for $x, y \in K$. In particular the group power x^k equals the multiple kx , so element orders are the same in both structures, and $xc = x + c$ whenever $c \in Z(K)$;
- (B3) every group automorphism of K is a Lie ring automorphism. This holds because $x + y = xy[x, y]^{-1/2}$ and the bracket are defined group-theoretically.

Let h also denote the automorphism $x \mapsto h x h^{-1}$ of K . It is a Lie ring automorphism of order 3 whose only fixed point is $0 = 1_K$. Hence $h - 1$ is injective, and therefore bijective, on the finite abelian group $(K, +)$. From $(h - 1)(h^2 + h + 1) = h^3 - 1 = 0$ we obtain $h^2 + h + 1 = 0$ in $\text{End}(K, +)$.

Let $7^e = \exp(K, +)$. The polynomial $t^2 + t + 1$ has the simple root 2 modulo 7, so by Hensel's lemma it has a root $\alpha \in \mathbb{Z}/7^e\mathbb{Z}$ with $\alpha \equiv 2 \pmod{7}$. Put $\beta = -1 - \alpha$. Then $t^2 + t + 1 = (t - \alpha)(t - \beta)$, $\alpha^2 = \beta$, $\alpha\beta = \alpha^3 = 1$, and $\alpha - \beta = 2\alpha + 1 \equiv 5 \pmod{7}$ is a unit. For $\gamma \in \{\alpha, \beta\}$ let $K_\gamma = \{x \in K : h(x) = \gamma x\}$. Since $(h - \alpha)(h - \beta) = 0$ and $\alpha - \beta$ is a unit, we have

$$K = K_\alpha \oplus K_\beta, \quad x = (\alpha - \beta)^{-1}((h - \beta)x - (h - \alpha)x).$$

For $x \in K_\gamma$ and $y \in K_\delta$ we have $h([x, y]) = \gamma\delta[x, y]$. Since h has no nonzero fixed points and $\alpha\beta = 1$, this gives

$$[K_\alpha, K_\beta] = 0 \quad \text{and} \quad [K_\alpha, K_\alpha] \subseteq K_{\alpha^2} = K_\beta.$$

Suppose that K has an element of order 49. Then K_α or K_β contains an element of additive order 49. Replacing h by h^{-1} , which interchanges K_α and K_β , we may assume that $y \in K_\alpha$ has order 49. By (6.1), $|B_G(y)| = 21$, so $B_G(y)$ is the subgroup of order 21 of the cyclic group $(\mathbb{Z}/49\mathbb{Z})^\times \cong C_{42}$. It therefore contains the element 8, which has order 7. Choose $g \in N_G(\langle y \rangle)$ with $g^{-1}yg = y^8$. The image of g in $B_G(y)$ has order 7, so it equals the image of the 7-part of g . Hence we may assume that g is a 7-element, i.e. $g \in K$. Then, by (B1) and (B2), and since $[y, g] \in [K, K] \leq Z(K)$,

$$8y = y^8 = g^{-1}yg = y[y, g] = y + [y, g], \quad \text{so} \quad [y, g] = 7y.$$

Writing $g = g_\alpha + g_\beta$ with $g_\gamma \in K_\gamma$, we get $[y, g] = [y, g_\alpha] + [y, g_\beta] = [y, g_\alpha] \in K_\beta$, while $7y \in K_\alpha$. Hence $7y \in K_\alpha \cap K_\beta = 0$, contradicting $o(y) = 49$. Therefore $\exp K = 7$.

In both cases every 7-element of G has order at most 7, and $O_5(G) = 1$ by Step 0. □

7. COMPUTATIONAL VERIFICATION

All computations were done with GAP 4.14.0 [GAP]; the code is reproduced in Appendix A. To test the cut property we used Proposition 2.1(ii) on a set of representatives x of the conjugacy classes. Write $n = o(x)$ and $b = |N_G(\langle x \rangle) : C_G(x)|$. Then x is inverse semi-rational if and only if $b = \varphi(n)$, or $2b = \varphi(n)$ and x^{-1} is not conjugate to x ; this is just the reformulation in Section 2, since $-1 \in B_G(x)$ if and only if x is real. The following results were obtained.

- (1) *Theorem A*. For $p \in \{2, 3, 5, 7\}$ we built $G_p \leq \text{Sym}(\mathbb{F}_p \times \mathbb{F}_p)$ from the four permutations $\pi(\delta_0)$, $\pi(d_s)$, $\pi(\tau_1)$ and $\pi(a_{s,0})$ of (4.2), with s a primitive root modulo p . In each case

we verified the following: $|G_p| = p^{p+1}(p-1)^2$; $O_p(G_p)$ has order p^{p+1} and exponent p^2 ; G_p equals the normalizer in S_{p^2} of $O_p(G_p)$, as computed by GAP; G_p is solvable; and every element of G_p is inverse semi-rational. For $p = 5$ (resp. $p = 7$) the group has 115 (resp. 3689) conjugacy classes. For $p = 5$ we also computed the character table: 83 irreducible characters are rational and the remaining 32 have field of values $\mathbb{Q}(i)$, in accordance with [Proposition 3.1](#) and [Lemma 4.2](#). This is also consistent with Ferraz's formula for the rank of the group of central units [[Fer04](#)]: in N_5 the number of real classes (unions of the conjugacy classes of x and x^{-1}) and the number of rational classes are both equal to 99.

- (2) [Proposition 5.2](#). For $p \in \{5, 7\}$ we ran over all 1700 groups G in the SmallGroups library of order $n \leq 2000$ with $p^3 \mid n$. For each one we tested whether G is solvable, whether $\exp O_p(G) > p$, and whether G is a cut group. No group satisfies all three conditions.
- (3) *Minimality within N_5* . For representatives K of the conjugacy classes of subgroups of a complement H of V in N_5 , the group $V \rtimes K$ is a cut group with $\exp O_5 > 5$ only for $K = H$.
- (4) [Proposition 5.4](#) for $p = 5$. The group $W_5 \leq S_{125}$ has order $5^{31} \cdot 4^3$ and equals $N_{S_{125}}(O_5(W_5))$, and $\exp O_5(W_5) = 125$. The stabilizer S of $\{(0, 0), (0, 1)\}$ in G_5 has order 5000 and is not a cut group: the element $x = \delta_1$ satisfies $|B_S(x)| = 2$.

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APPENDIX A. GAP CODE

The following file reproduces all computations of [Section 7](#). In our run it finished in about seven minutes on a two-core virtual machine that was also running other jobs.

```
# GAP 4.14 code accompanying "A negative answer to a question of Baechle
# on solvable cut groups". Run with: gap -o 8g bachle_q3_verify.g

# -----
# Cut test (Proposition 2.1(ii)): x is inverse semi-rational iff
# |B_G(x)| = phi(o(x)), or |B_G(x)| = phi(o(x))/2 and x is not real.
IsISR := function(G, x, cc)
  local o, b;
  o := Order(x);
  b := Size(Normalizer(G, Subgroup(G, [x]))) / Size(Centralizer(G, x));
  return b = Phi(o) or (2*b = Phi(o) and not x^-1 in cc);
end;;

IsCutGroup := function(G)
  return ForAll(ConjugacyClasses(G), cc -> IsISR(G, Representative(cc), cc));
end;;

# -----
# The group G_p acting on Omega = F_p x F_p, formula (4.2) of the paper:
# f : (i,j) -> (i, j + f(i)), d_s : (i,j) -> (i, s j), a : (i,j) -> (a(i), j)
BuildG := function(p)
  local pts, idx, perm, s;
  pts := Cartesian([0..p-1], [0..p-1]);
  idx := function(i, j) return Position(pts, [i mod p, j mod p]); end;
  perm := f -> PermList(List(pts, q -> CallFuncList(idx, f(q[1], q[2]))));
  s := IntFFE(Z(p)); # a primitive root mod p
  return Group(
    perm(function(i,j) if i = 0 then return [i, j+1]; fi; return [i, j]; end), # delta_0 in V
    perm({i,j} -> [i, s*j]), # d_s in D
    perm({i,j} -> [i+1, j]), # tau_1 in T
    perm({i,j} -> [s*i, j])); # a_{s,0} in A
end;;

CheckG := function(p)
  local G, P;
  G := BuildG(p); P := PCore(G, p);
  Print("p = ", p, ": |G_p| = ", Size(G), " (= p^(p+1)(p-1)^2: ", Size(G) = p^(p+1)*(p-1)^2,
    "\n");
  Print(" |O_p| = p^n", Log(Size(P), p), ", exp O_p = ", Exponent(P),
    ", O_p Sylow in S_{p^2}: ", Size(P) = p^(p+1), "\n");
  Print(" G_p = N_{S_{p^2}}(O_p(G_p)): ", G = Normalizer(SymmetricGroup(p^2), P), "\n");
  Print(" solvable: ", IsSolvableGroup(G), ", classes: ", NrConjugacyClasses(G),
    ", cut: ", IsCutGroup(G), "\n");
```

```

    return G;
end;;

for p in [2, 3, 5, 7] do CheckG(p); od;

# Character fields of G_5
T := CharacterTable(BuildG(5));;
Print("G_5: ", Length(Irr(T)), " irreducible characters; fields: ",
      Collected(List(Irr(T), chi -> Field(Rationals, List(chi)))), "\n");

# -----
# Remark 5.3: subgroups V x| K of G_5 (K <= H up to conjugacy)
SubgroupCheck := function(p)
  local S, P, G, V, H, K, GK, res;
  S := SymmetricGroup(p^2); P := SylowSubgroup(S, p); G := Normalizer(S, P);
  V := First(NormalSubgroups(P), N -> IsElementaryAbelian(N) and Size(N) = p^p and IsNormal(G,
    N));
  H := ComplementClassesRepresentatives(G, V)[1];
  res := [];
  for K in List(ConjugacyClassesSubgroups(H), Representative) do
    GK := ClosureGroup(V, K);
    if Exponent(PCore(GK, p)) > p and IsCutGroup(GK) then Add(res, Size(K)); fi;
  od;
  Print("p = ", p, ": orders of K <= H with V x| K cut and exp 0_p > p: ", res,
        " (|H| = ", Size(H), ")\n");
end;;
SubgroupCheck(5);

# -----
# Proposition 5.4 for p = 5: W_5 = F^Omega x| (D_3 x G_5) acting on Omega x F (125 points)
p := 5;; s := IntFFE(Z(p));; G5 := BuildG(p);;
pts := Cartesian([0..p-1], [0..p-1]);;
idx := function(i, j) return Position(pts, [i mod p, j mod p]); end;;
S := Stabilizer(G5, [idx(0,0), idx(0,1)], OnSets);; # stabilizer of phi = delta_(0,0) +
  delta_(0,1)
x := PermList(List(pts, function(q) if q[1] = 1 then return idx(q[1], q[2]+1); fi;
  return idx(q[1], q[2]); end));; # the base element
  delta_1
Print("S = Stab_{G_5}({(0,0),(0,1)}): |S| = ", Size(S), ", cut: ", IsCutGroup(S),
      "; x in S: ", x in S, ", |B_S(x)| = ",
      Size(Normalizer(S, Subgroup(S, [x]))) / Size(Centralizer(S, x)), "\n");
pts3 := Cartesian([0..p-1], [0..p-1], [0..p-1]);;
idx3 := function(a, b, c) return Position(pts3, [a mod p, b mod p, c mod p]); end;;
perm3 := f -> PermList(List(pts3, q -> CallFuncList(idx3, f(q[1], q[2], q[3]))));;
lift := g -> perm3(function(a, b, c) local q; q := pts[idx(a, b)^g]; return [q[1], q[2], c];
  end);;
W := Group(Concatenation(List(GeneratorsOfGroup(G5), lift),
  [ perm3(function(a, b, c) if a = 0 and b = 0 then return [a, b, c+1]; fi; return [a, b,
    c]; end),
    perm3({a, b, c} -> [a, b, s*c]) ]));;
P3 := PCore(W, p);;
Print("W_5: |W_5| = 5^31*4^3: ", Size(W) = 5^31*4^3, ", |O_5(W_5)| = 5^", Log(Size(P3), 5),
      ", exp 0_5(W_5) = ", Exponent(P3), ", W_5 = N_{S_125}(O_5(W_5)): ",
      W = Normalizer(SymmetricGroup(125), P3), "\n");

# -----
# Proposition 5.2: solvable groups of order <= 2000 with p^3 | n, p in {5,7}

```

```
found := [];; nr := 0;;
for p in [5, 7] do
  for n in Filtered([1..2000], n -> n mod p^3 = 0) do
    for i in [1..NrSmallGroups(n)] do
      nr := nr + 1;
      G := SmallGroup(n, i);
      if IsSolvableGroup(G) and Exponent(PCore(G, p)) > p and IsCutGroup(G) then
        Add(found, [n, i, p]);
      fi;
    od;
  od;
od;
Print("SmallGroups search: ", nr, " groups examined; solvable cut groups with exp 0_p > p: ",
      found, "\n");
QUIT;
```